IAU STRATEGIC PLAN ASTRONOMY FOR DEVELOPMENT

GEORGE MILEY
Leiden University

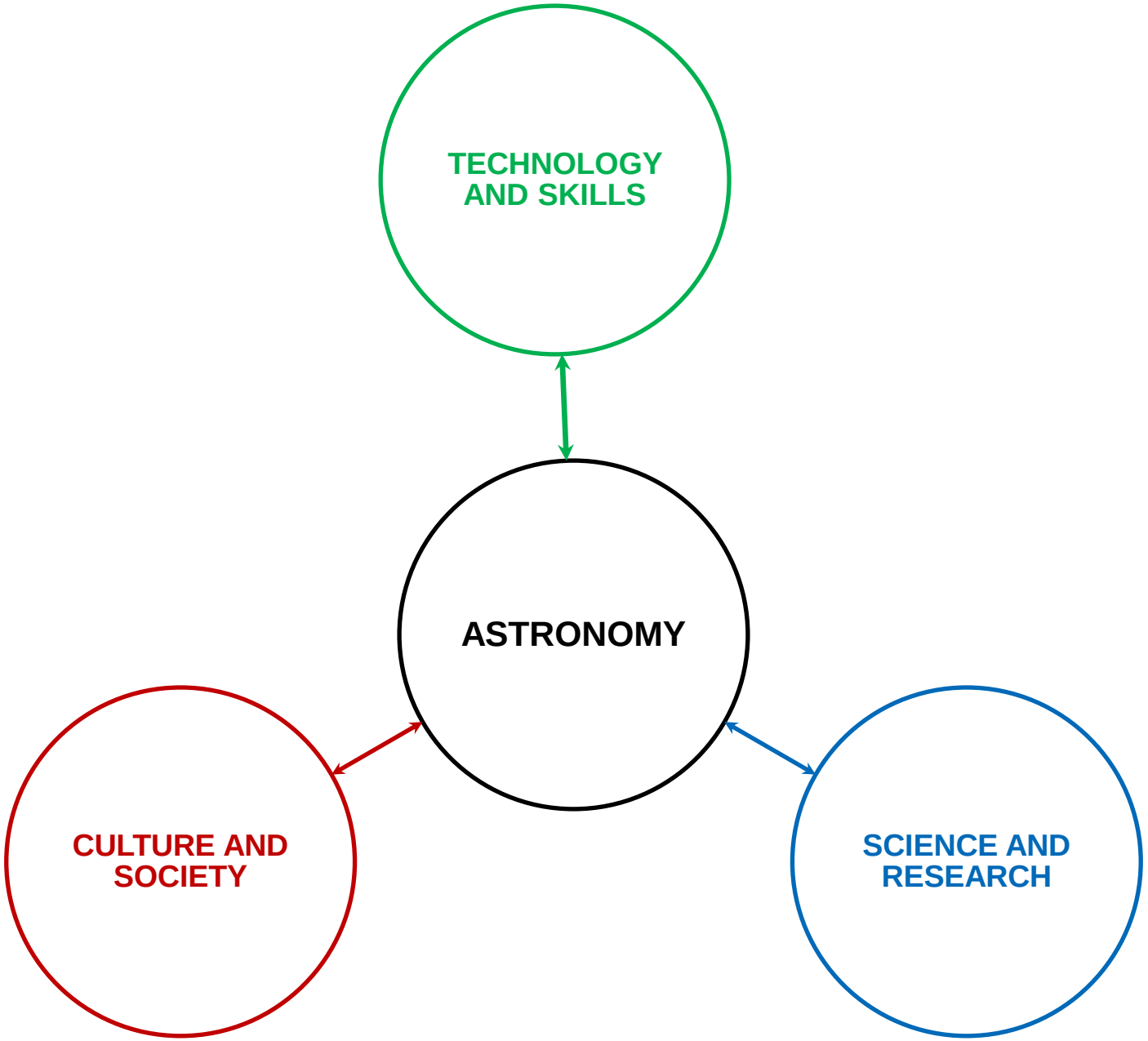
Emeritus IAU Vice President
(Portfolio: Development & Education)



- **Rationale and motivation**
- **Strategy**
- **3. Young children and astronomy for development**
- **Implementation**
  - **How far have we got?**
    - **Kevin Govender**

ftp://ftp.saao.ac.za/outgoing/kg/astro4dev/stratplan_2012update.pdf

THREE PILLARS OF DEVELOPMENT



TECHNOLOGY DRIVEN BY ASTRONOMY

Charge coupled devices (cameras)

Wireless Internet

Most accurate clocks

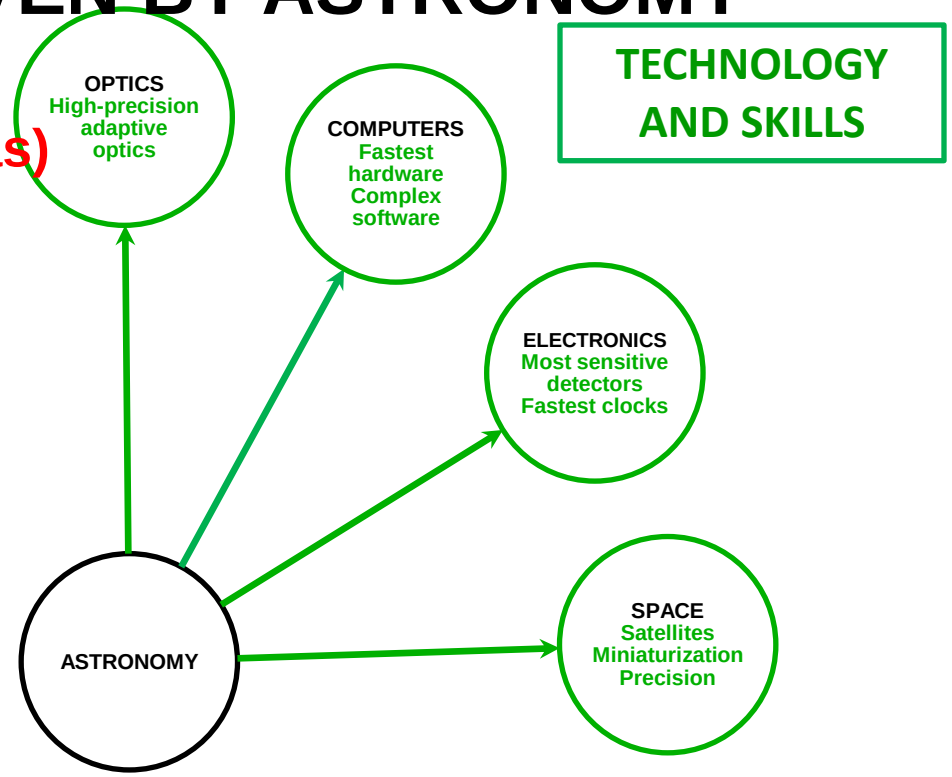
Sensitive antennas

Low-noise amplifiers

Supercomputers

GPS navigation

Medical imaging

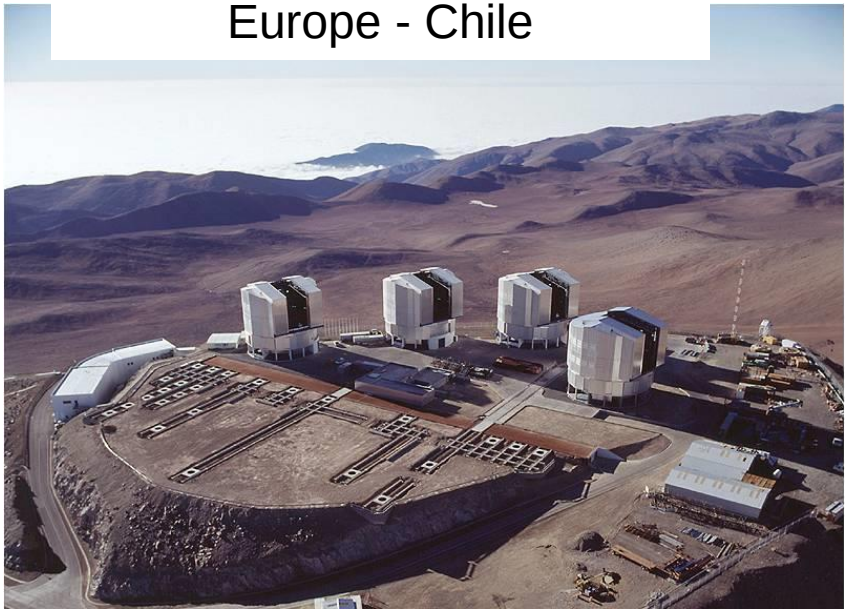


NEED TO OBSERVE FAINTEST OBJECTS

CUTTING-EDGE TECHNOLOGY

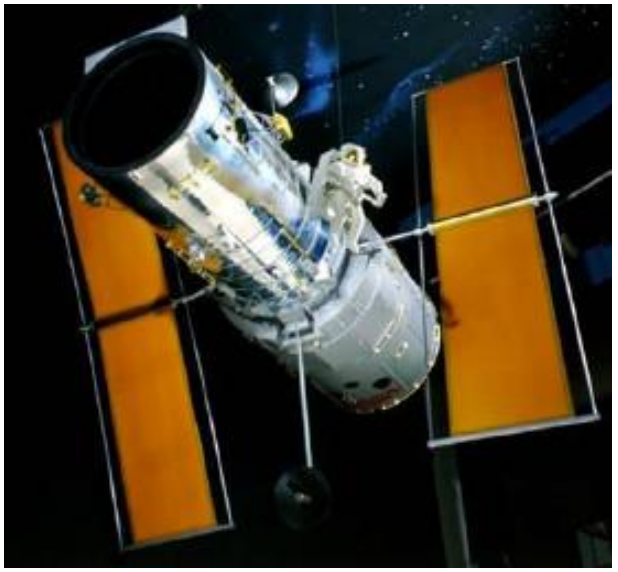
VERY LARGE TELESCOPE

Europe - Chile



HUBBLE SPACE TELESCOPE

US - Europe



SUBARU – Japan/ Hawaii



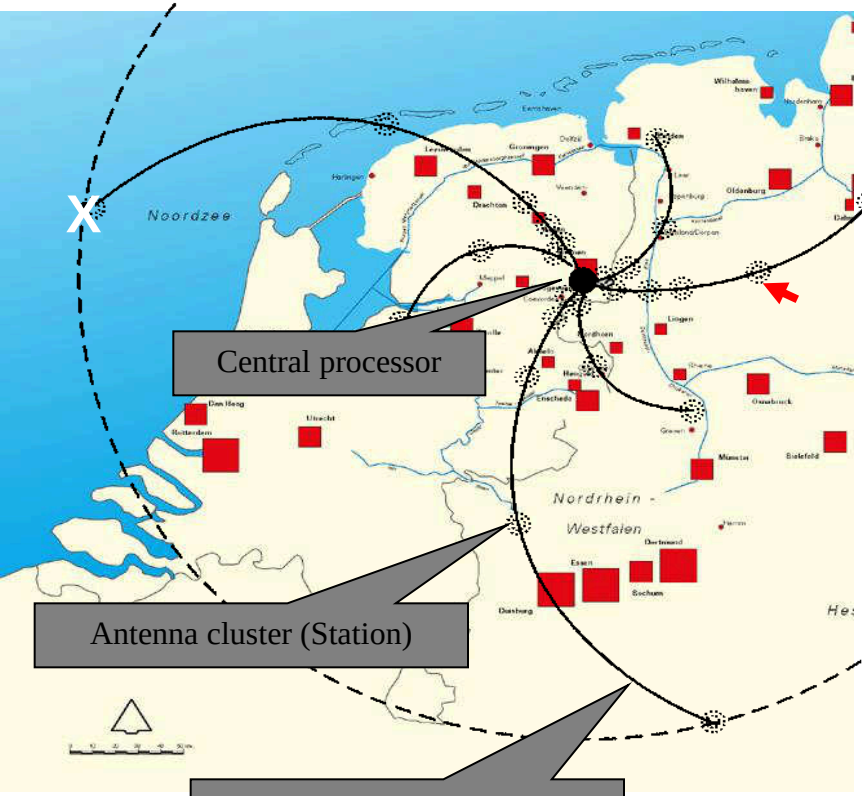
SALT – Southern Africa +



ASTRONOMY DRIVES OTHER APPLICATIONS – EXAMPLE 1

GEOPHYSICS PIGGYBACKS ON RADIO ASTRONOMY

LOFAR ARRAY



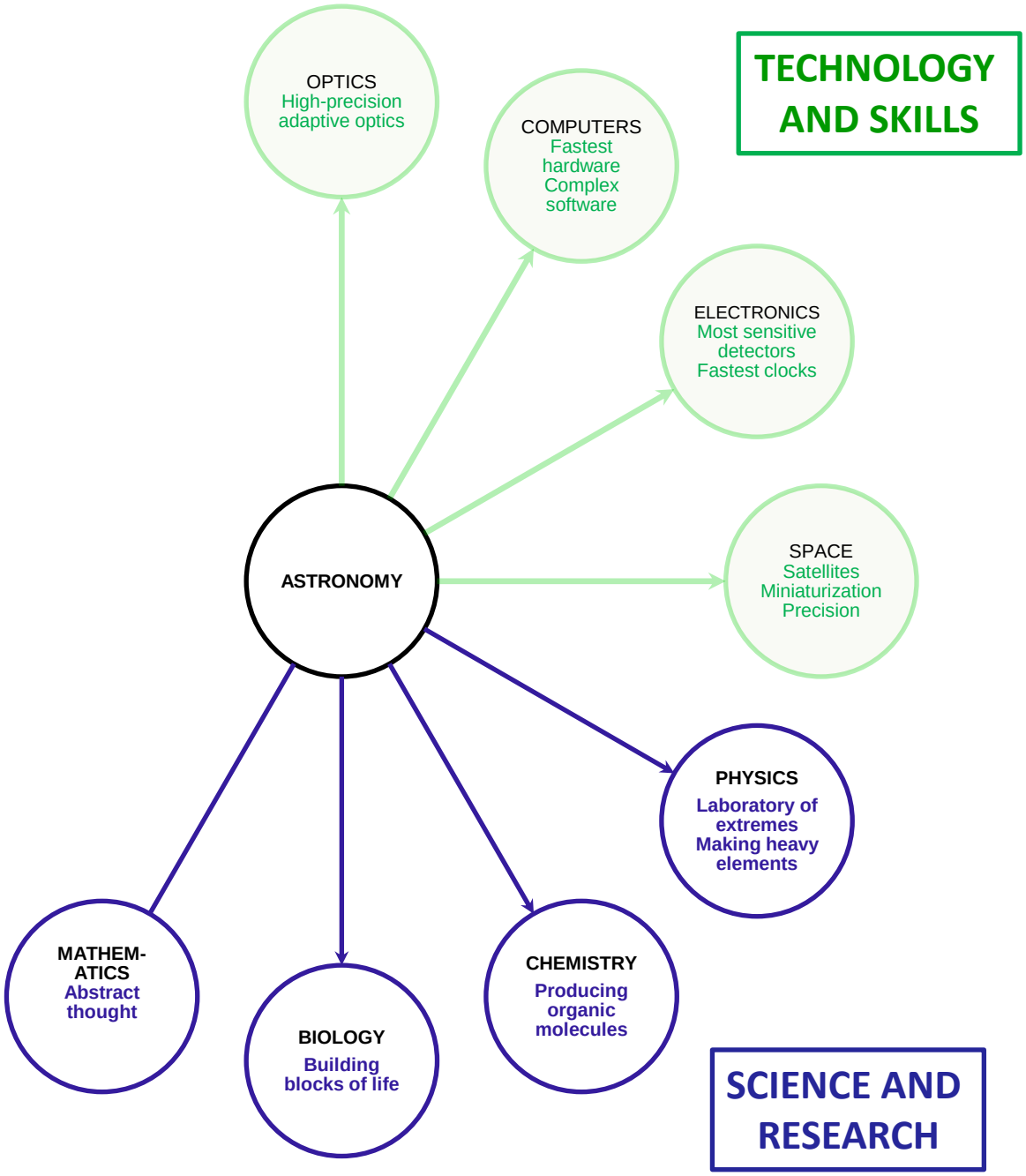
~5000 primitive radio dipoles
Steered by 30TFlop supercomputer



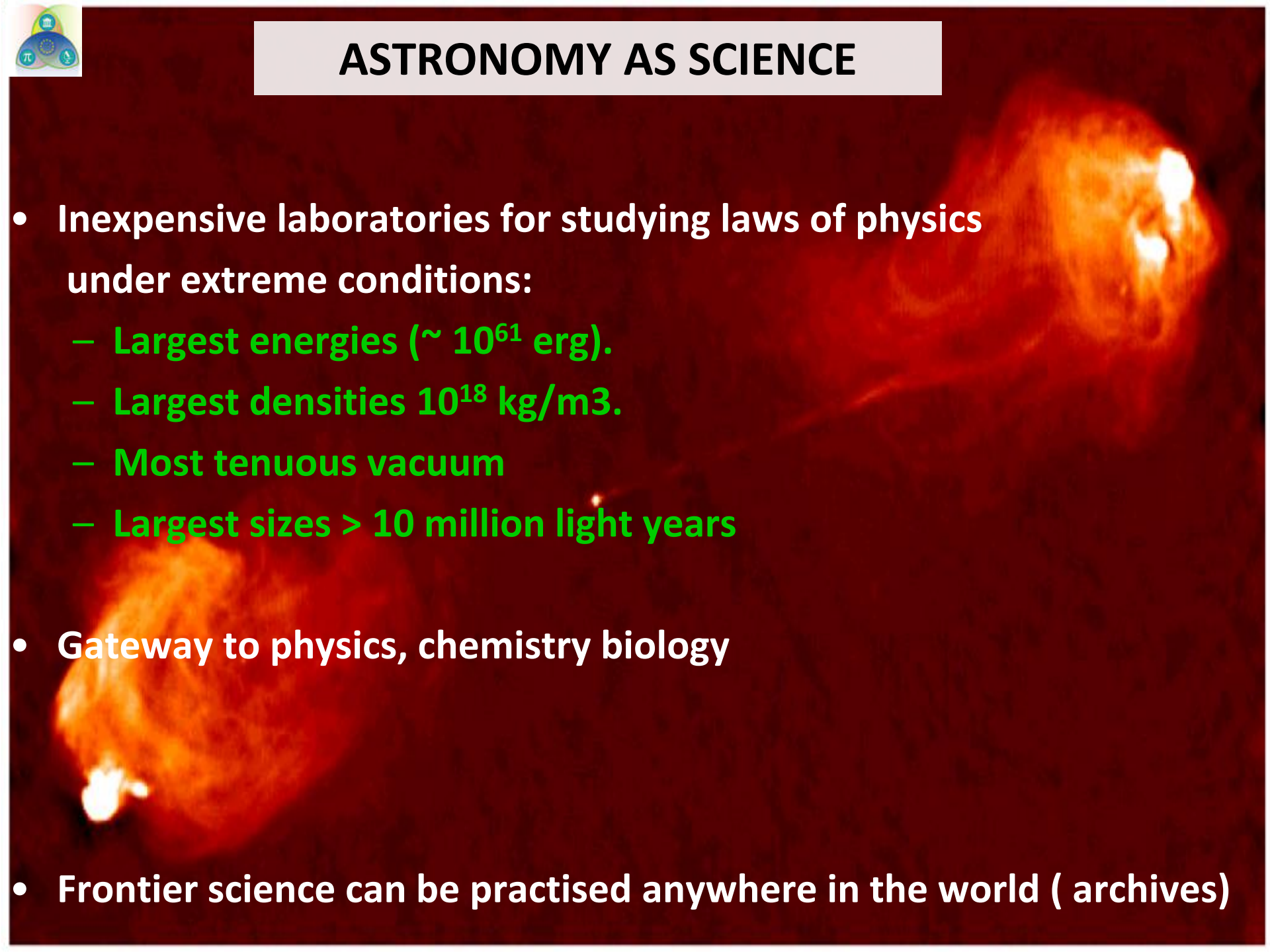
Array of geophones to measure earth subsistence due to gas mining



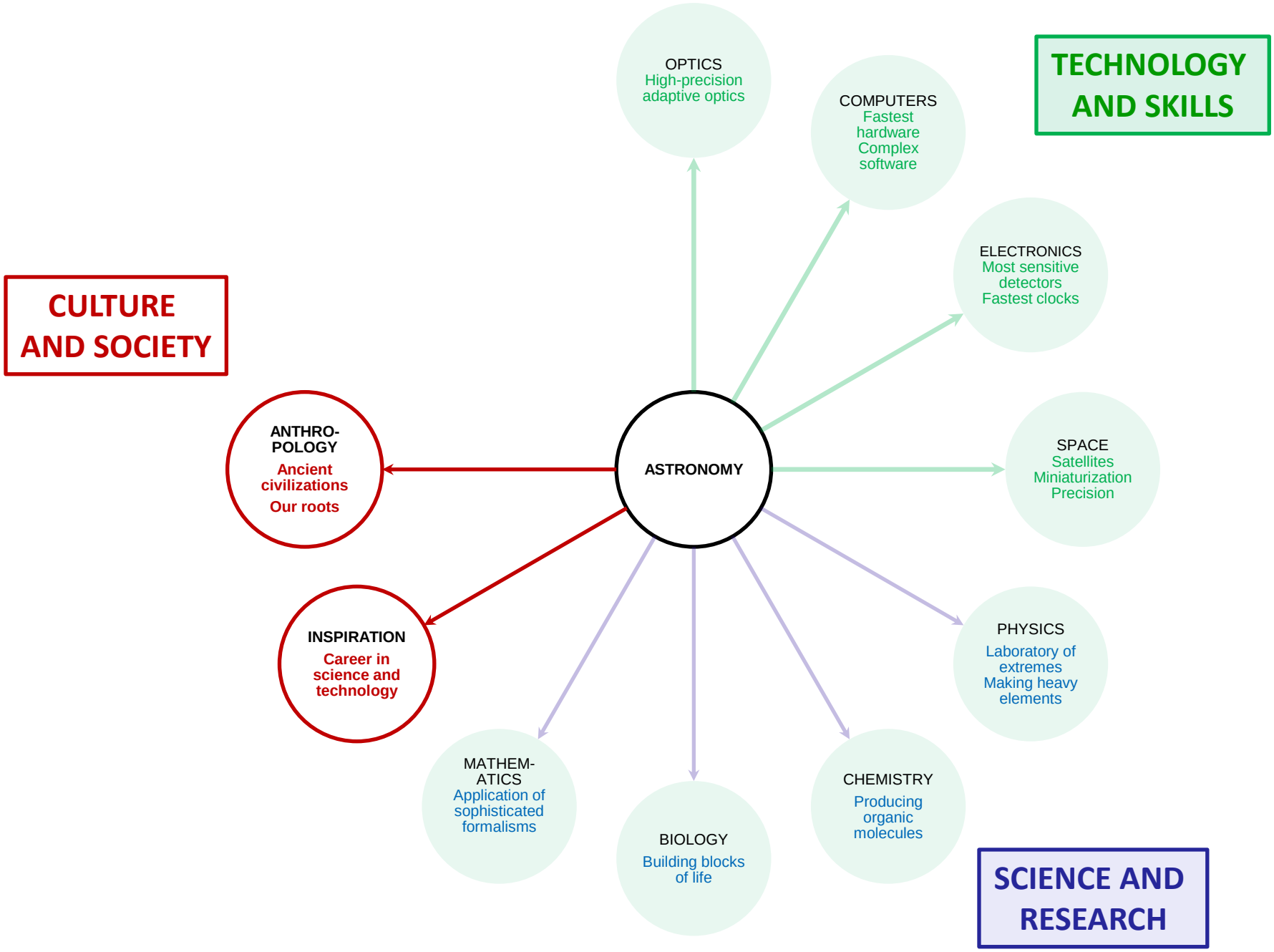



TECHNOLOGY AND SKILLS
OPTICS
High-precision adaptive optics
COMPUTERS
Fastest hardware
Complex software
ELECTRONICS
Most sensitive detectors
Fastest clocks
SPACE
Satellites
Miniaturization
Precision
ASTRONOMY
PHYSICS
Laboratory of extremes
Making heavy elements
MATHEM-ATICS
Abstract thought
BIOLOGY
Building blocks of life
CHEMISTRY
Producing organic molecules
SCIENCE AND RESEARCH

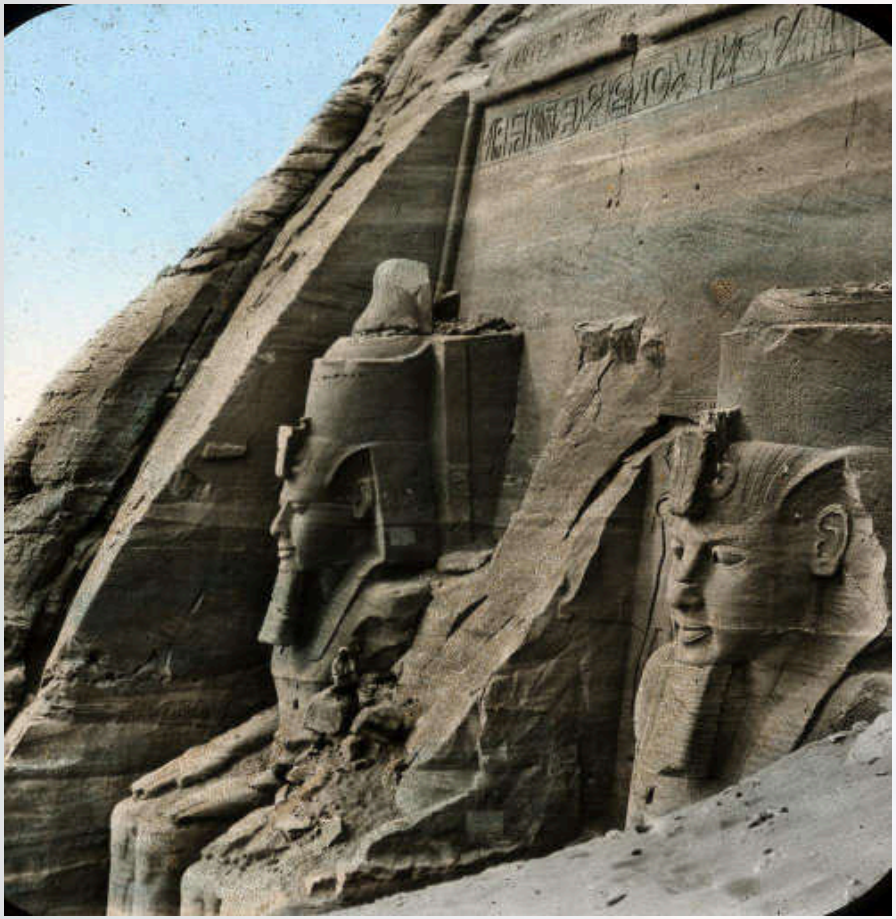
ASTRONOMY AS SCIENCE



- Inexpensive laboratories for studying laws of physics under extreme conditions:
  - Largest energies (~ 10^{61} erg).
  - Largest densities 10^{18} kg/m3.
  - Most tenuous vacuum
  - Largest sizes > 10 million light years
- Gateway to physics, chemistry biology
- Frontier science can be practised anywhere in the world (archives)


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Application of sophisticated formalisms
ASTRONOMY
CULTURE AND SOCIETY
ANTHRO-POLOGY
Ancient civilizations
Our roots
INSPIRATION
Career in science and technology
SCIENCE AND RESEARCH

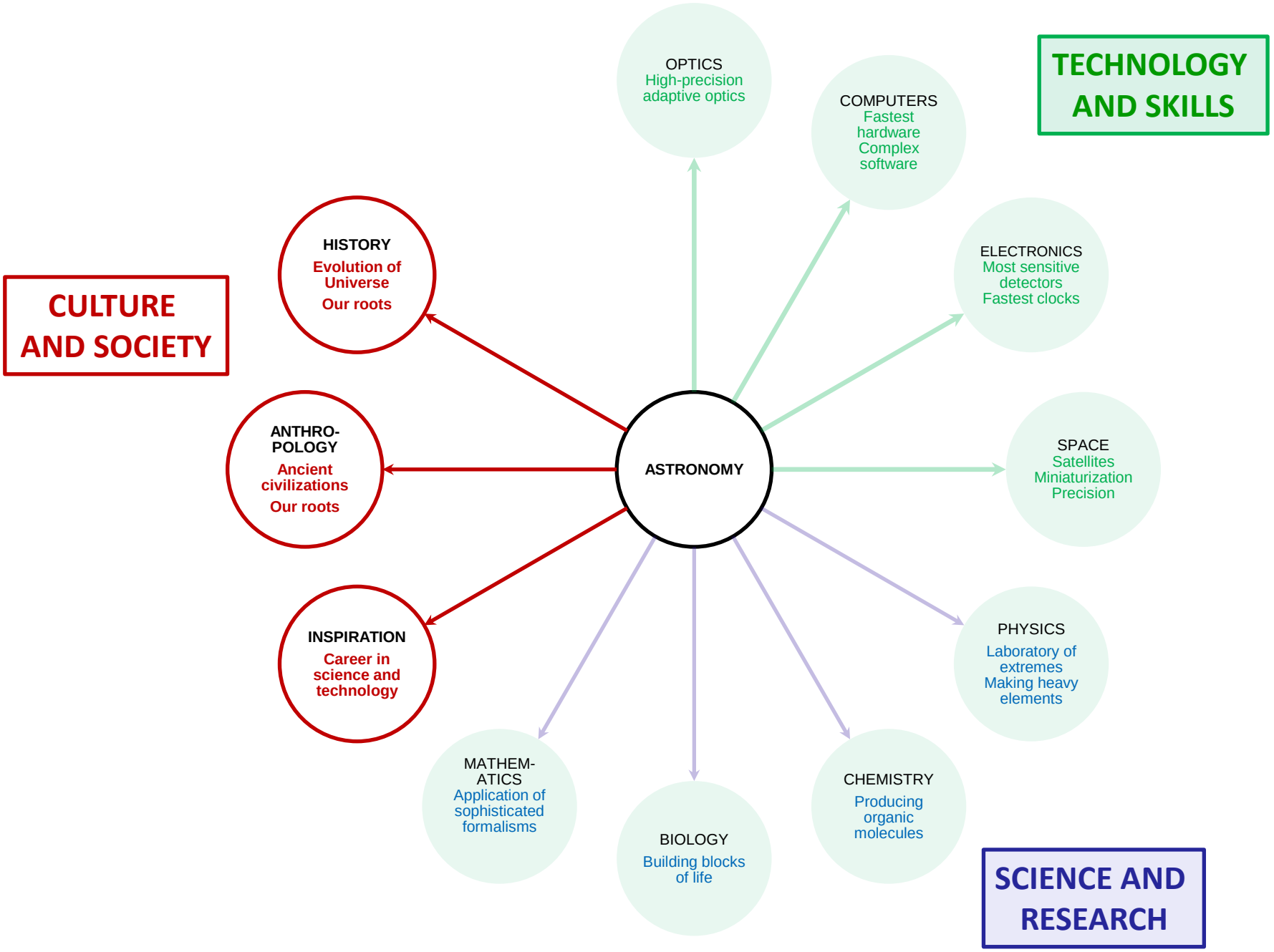
MIDDLE EAST AND NORTH AFRICA HAVE RICH ASTRONOMICAL HERITAGE

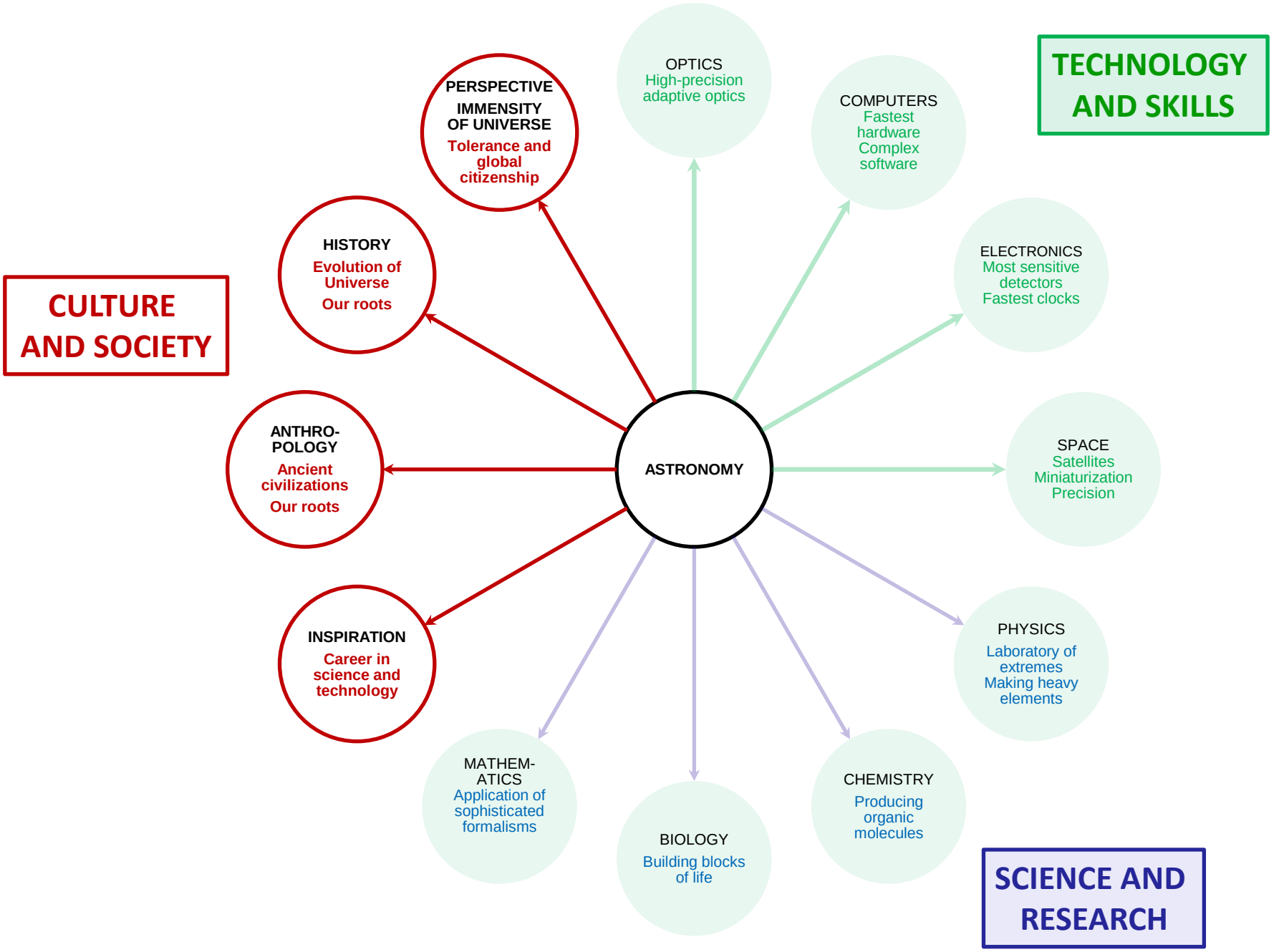


EGYPT - ABU SIMBEL
Sun illuminates sculpture on back wall

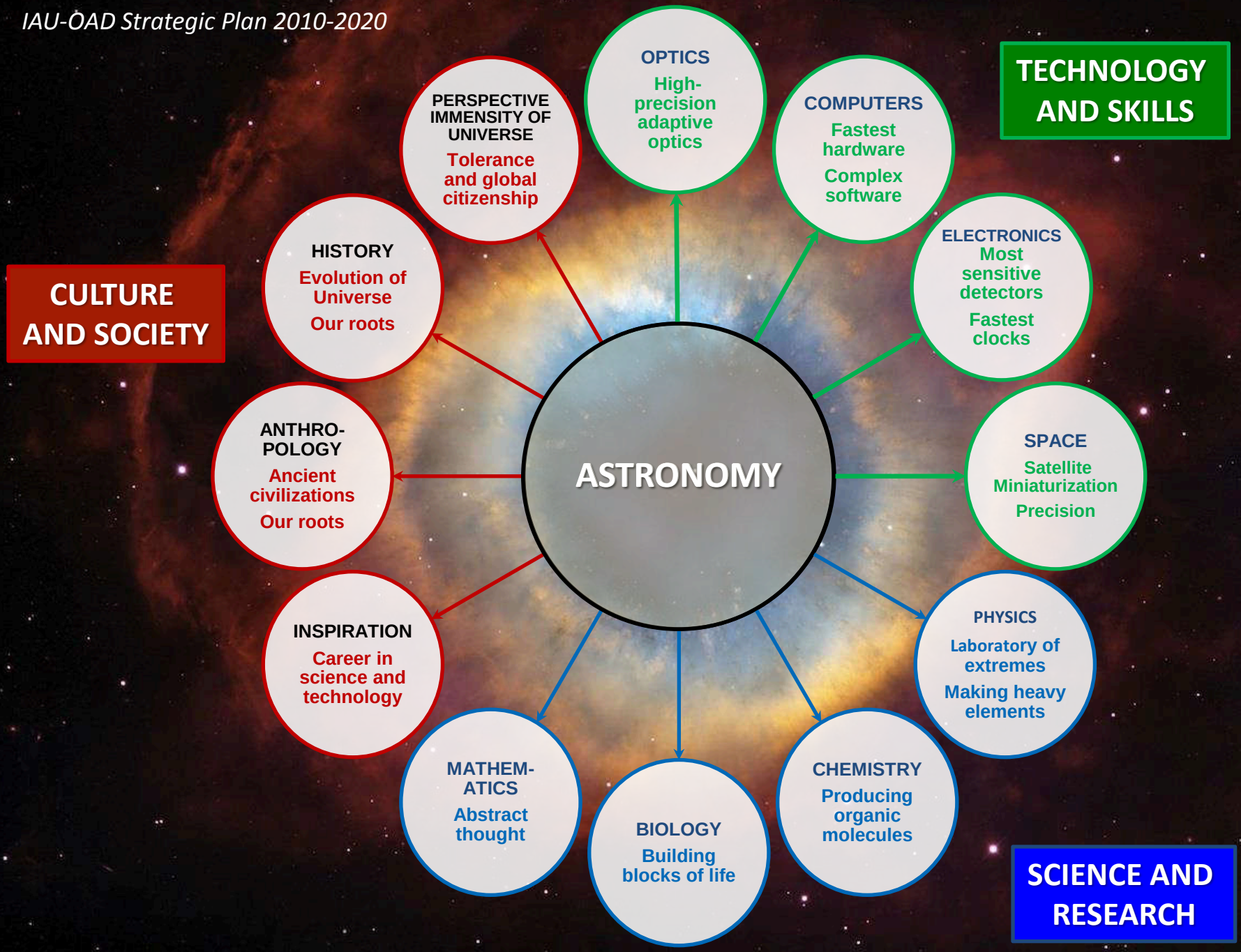


JORDAN - PETRA
At winter solstice
sun illuminates ancient deity
and mountain casts shadow of lion' head


CULTURE AND SOCIETY
HISTORY
Evolution of Universe
Our roots
ANTHRO-POLOGY
Ancient civilizations
Our roots
INSPIRATION
Career in science and technology
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SCIENCE AND RESEARCH


CULTURE AND SOCIETY
PERSPECTIVE
IMMENSITY OF UNIVERSE
Tolerance and global citizenship
HISTORY
Evolution of Universe
Our roots
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IAU-OAD Strategic Plan 2010-2020



IAU STRATEGIC PLAN
"ASTRONOMY FOR DEVELOPMENT"

http://iau.org/static/education/strategicplan_091001.pdf

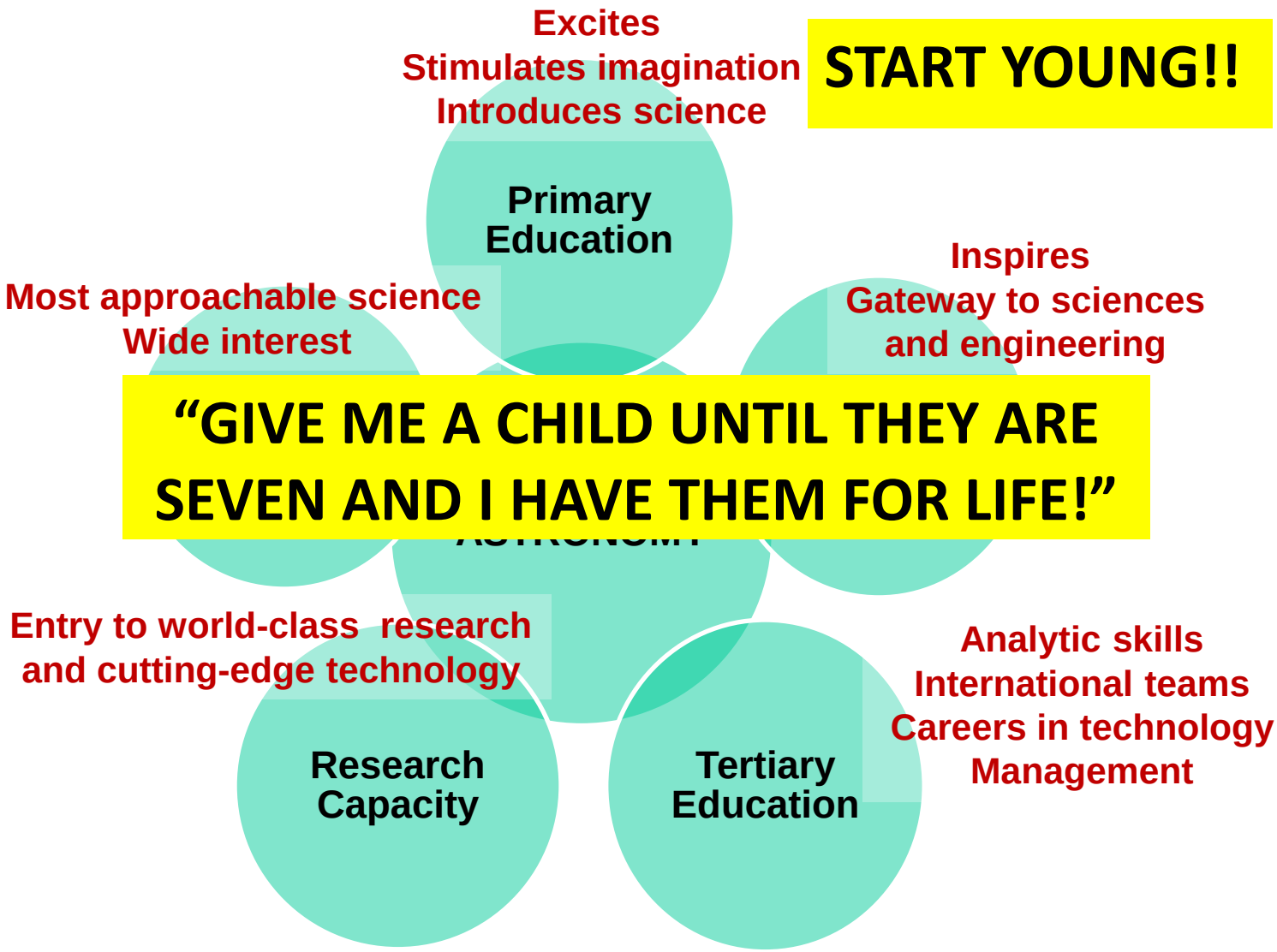
- Long term vision and goals for decade

- Strategy
  - Integrated phased approach
  - Increased regional involvement (bottom-up)
    - Demand driven from the regions
  - Enlarge no, of volunteers
    - Multidisciplinary, Use expatriates
  - Build on IYA2009

- Implementation roadmap includes
  - Set up and operate global coordinating office
    - OAD
  - Regional coordinating offices
    - ROADs



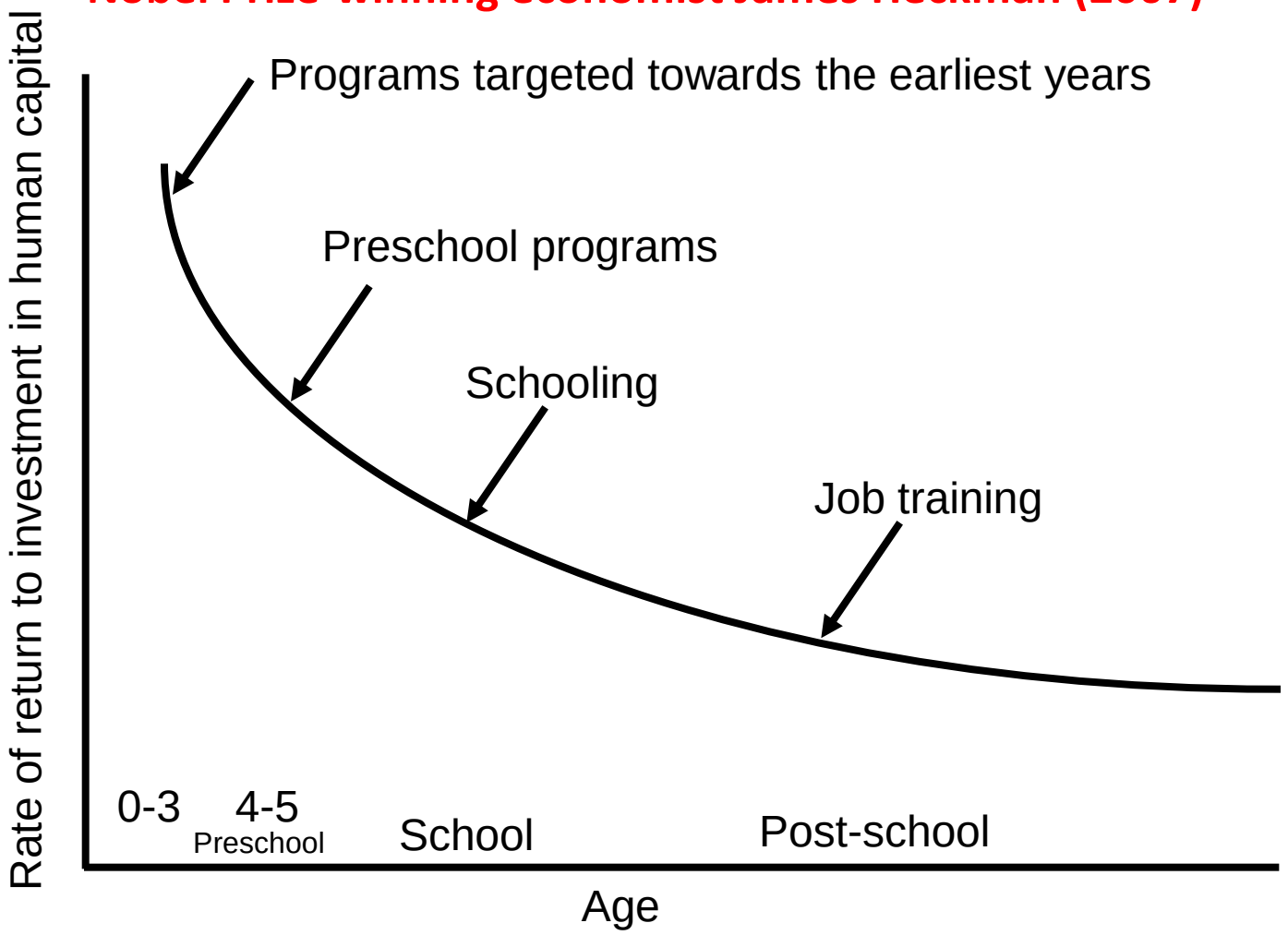
TOOLS OF ASTRONOMY FOR DEVELOPMENT



CAPACITY BUILDING – START YOUNG

Rates of Return for Human Capital Investment at Different Ages

Nobel Prize-winning economist James Heckman (2007)

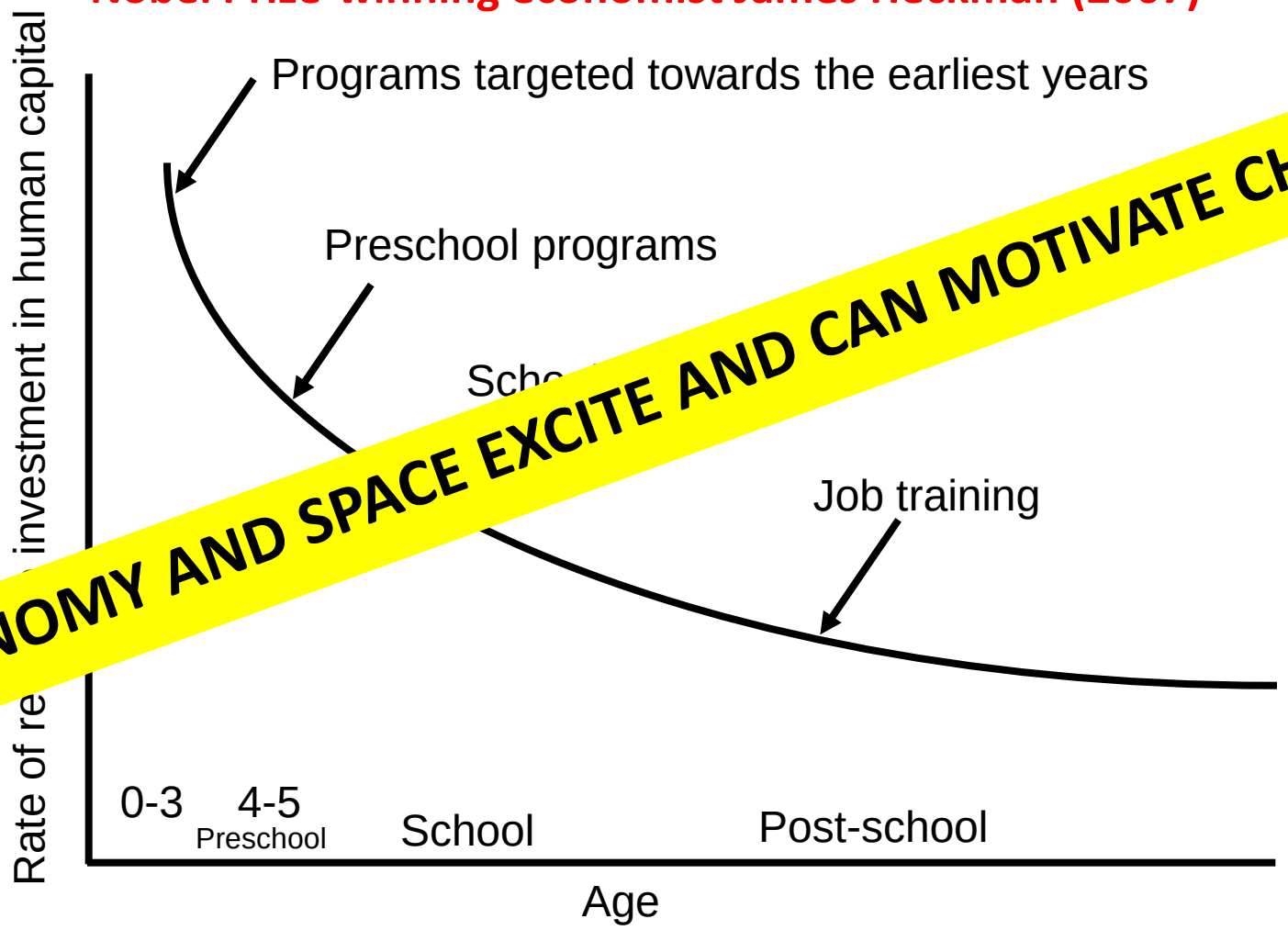


"Strong case for funding interventions in early childhood for disadvantaged children"

CAPACITY BUILDING – START YOUNG

Rates of Return for Human Capital Investment at Different Ages

Nobel Prize-winning economist James Heckman (2007)



ASTRONOMY AND SPACE EXCITE AND CAN MOTIVATE CHILDREN

"Strong case for funding interventions in early childhood for disadvantaged children"

MUCH MORE THAN JUST ECONOMIC EFFECTIVENESS
AWARENESS OF UNIVERSE GIVES PERSPECTIVE
CAN HELP COMBAT FANATICISM

"*Fanatic ethnic, religious or national identifications* are difficult to support when we see our planet as a fragile, blue crescent fading to become an inconspicuous point of light against the bastion and citadel of the stars. "
CARL SAGAN



UNIVERSE AWARENESS (UNAWE)
Astronomy-based program with social goals



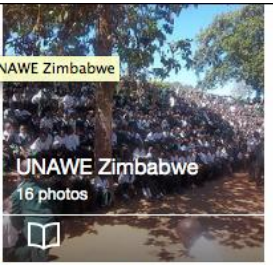
Exposes DISADVANTAGED young children (4 – 10) to INSPIRATIONAL aspects of astronomy

- Use INSPIRATION and FUN of astronomy to
  - Introduce excitement of science
    - Demonstrate power of rational thought
    - Motivate development of language and numeric skills

- Use PERSPECTIVE of astronomy to
  - Broaden children's minds
    - Stimulate sense of internationalism, tolerance, respect at formative age

- Global reach
  - Activities in 63 countries
    - 5th UNAWE International Workshop Leiden, 5 – 9 October 2015

UNAWE IN MORE THAN 60 COUNTRIES







































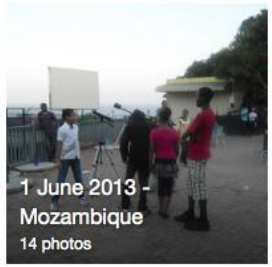
































UNAWE RESOURCES: e.g. EARTH BALL WITHOUT BORDERS



- > 10 000 distributed to schools in > 60 countries

UNAWE RESOURCES: UNIVERSE IN A BOX





- ~ 1000 distributed to schools over whole world

ISLAMIC ASTRONOMY HERITAGE TOOLBOX

EU Space Awareness deliverable : Prototype ~ June 2016

- **Development**
  - **Dr. Cecilia Scorza, Max Planck Heidelberg/ Heidelberg University**
    Prof. Hassane Darhmaoui, Al Akhawayn University, Ifrane, Morocco

- **Audience**
  - **Migrant and ethnic European children and their teachers**

- **Content**
  - **Stories, hands-on activities, teacher manual etc.**

- **Philosophy – "Explore the journey of ideas that grow and influence each other"**
  - **Shared history – of collaboration, coexistence and tolerance**
  - **Shared night sky - Arabic names of stars**
  - **Need for calendars**
    - **Demonstrate influence of Islamic civilisation on modern space science**
    - **Combat stereotypes**

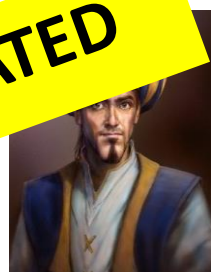
ISLAMIC ASTRONOMY HERITAGE TOOLBOX

Based on 4 Moslim scientists – 2 women en 2 men









INPUT AND FEEDBACK WOULD BE APPRECIATED

Fatima Al Fihri

9th century, Fez Morocco

Founder of Qarawiyyin University, Fez

Abi,

10th century, Aleppo, Syria.

She was famous scientist who designed and constructed astrolabes.

Abd al-Rahman al-Sufi

(902-960), Rey, Iran.

Identified Large Magellanic Cloud and made earliest recorded observation of Andromeda galaxy.

Ibn Al Haytham

(Alhazen, 965-1040), Basra, Iraq.

Among most influential scientists of all time. Referred to as father of experimental physics, modern optics and scientific methodology.

TUNISIAN ASTRO-BUS





- Pioneering activity of La Cité des Sciences, Tunis.
- Transports small telescope + mini-planetarium + exhibition.
- Inspiring children throughout Tunisia, even in remotest villages.
- During 2008 > reached 150,000 children!

Idea could be exported to many countries
Also implemented by Galileo –Mobile

POLITICAL IMPORTANCE - ASTRONOMY FOR DEVELOPMENT

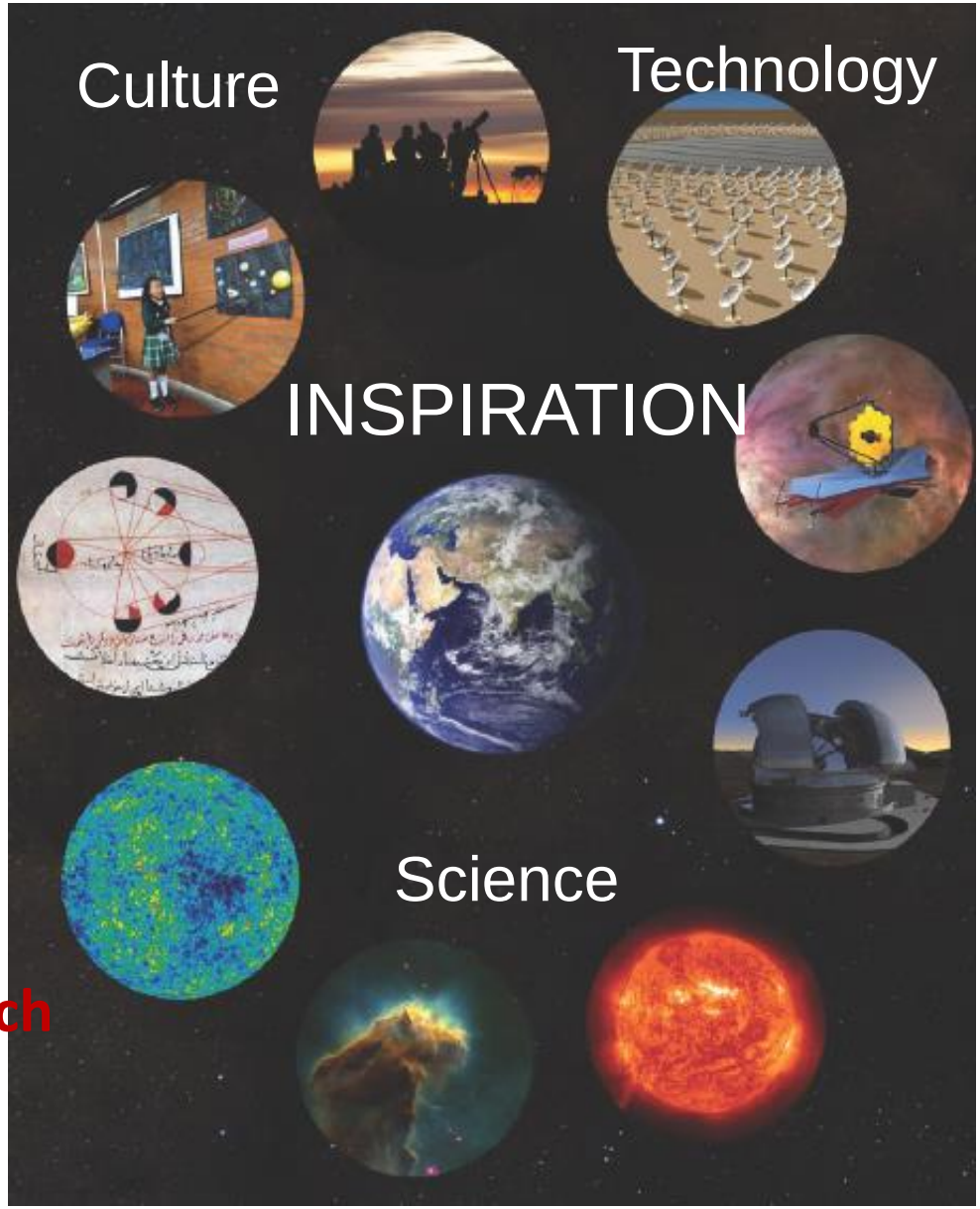
- **EUROPEAN PARLIAMENT; Written Declaration 45/2011 March 2012 signed by 394 MEPs**
  - SUPPORTS ***THE DEVELOPMENT OF SCIENCE CAPACITY*** *in Africa through greater investment in research infrastructures,* ***WITH PARTICULAR FOCUS ON RADIO ASTRONOMY***

- **AFRICAN UNION; Declaration AU Assembly 18th ordinary session 29 -30 Jan 2012**
  - PROPOSES the inclusion of ***RADIO ASTRONOMY AS A PRIORITY FOCUS AREA FOR AFRICA'S INTERNATIONAL SCIENCE AND TECHNOLOGY PARTNERSHIPS***

- **PR CHINA; President Xi, 21 August 2012 at IAU GA**
  - ***ASTRONOMY IS A CRUCIAL FIELD OF BASIC RESEARCH. We will make larger and larger investments in such a field…. that will benefit humankind.***
  - ***PUBLIC OUTREACH SHOULD BE GIVEN EQUAL EMPHASIS AS SCIENTIFIC RESEARCH…… to inspire the creativity for science and technological innovation among the public.***

FINAL REMARKS

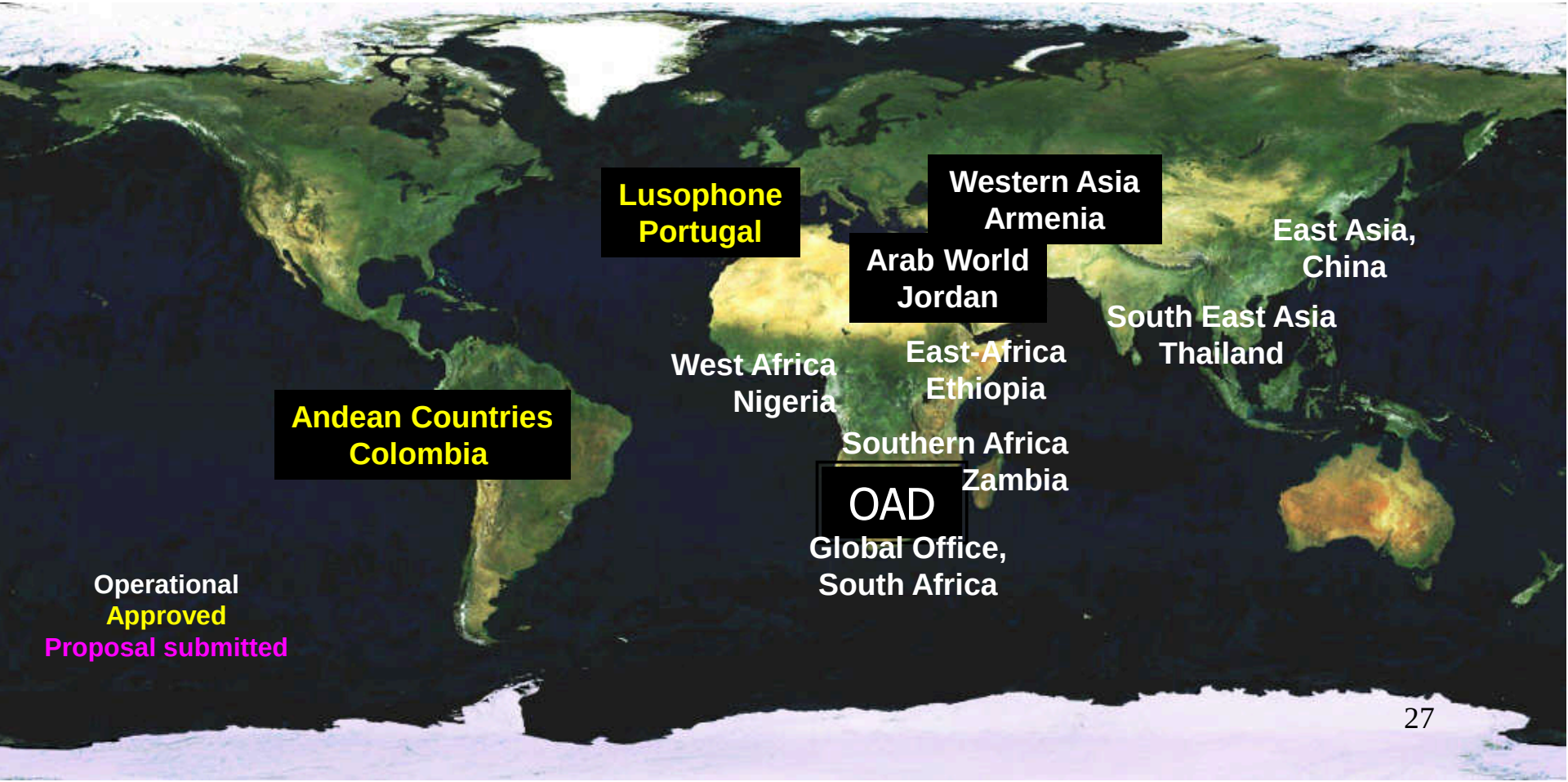
- **Astronomy FOR Development**
  - **Mutually beneficial for society AND astronomy**
  - **Unique for an international scientific organisation**
  - **Sets example for other curiosity-driven sciences**
  - **Needs joint effort by professional and amateur astronomers, educators/teachers, outreach experts**
    - **Equal partnership**





INAUGURATION
IAU ARAB REGIONAL OFFICE OF ASTRONOMY FOR DEVELOPMENT

CONGRATULATIONS DR. IR. AWNI KHASAWNEH



NALEDI PANDOR
SOUTH AFRICAN MINISTER OF SCIENCE AND TECHNOLOGY
ADDRESSING SA BUSINESS LEADERS



UNAWE RESOURCES: UNIVERSE IN A BOX



Winner of Scientix Prize 2015

Needs localisation and translation into Arabic



- **~ 1000 distributed to schools over whole world**